SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

Interdisciplinary Project Report on

"Green Synthesis of TiO_2 nanoparticles and development of PLA-TiO_2 nanocomposite for electrical insulator applications"

submitted in partial fulfillment for the completion of II Semester

BACHELOR OF ENGINEERING

Submitted by

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

2025-26

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CERTIFICATE

Certified that the project work entitled "Green synthesis of TiO_2 nanoparticles and development of PLA-TiO_2 nanocomposite for electrical insulator applications" is a bonafide work carried out by ESHWARA CHANDRA P N (1SI25EE010), LIKHITHA R (1SI25BT-008), MOKSHITH D (1SI25IM043), POORVI CHOUKIMATH (1SI25IS069), PRAHARSH TANDON (1SI25CS169) and SATYA PRAKASH (1SI25CI045) in partial fulfillment for the completion of II Semester of Bachelor of Engineering from Siddaganga Institute of Technology, an autonomous institute under Visvesvaraya Technological University, Belagavi during the academic year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the completion of II Semester of Bachelor of Engineering.

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Course Outcomes

After successful completion of Interdisciplinary project, students will be able to

CO1: To identify a problem through literature survey, knowledge of contemporary engineering technology and formulate the engineering problem.

CO2: To provide sustainable engineering solution considering health, safety, legal, cultural issues and also demonstrate concern for environment.

CO3: To identify and apply the mathematical concepts, science concepts, engineering and management concepts necessary to implement the identified engineering problem

CO4: To select the engineering tools/components required to implement the proposed solution for the identified engineering problem.

CO5: To demonstrate effective written communication through the project report and to engage in effective oral communication through power point presentation and demonstration of the project work.

CO6: To perform in the team, contribute to the team and mentor/lead the team.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO-1		2									2		
CO-2						2							
CO-3		2											
CO-4					3								
CO-5							2		3				
CO-6								3					
Average		2			3	2	2	3	3		2		

Attainment level: - 1: Slight (low) 2: Moderate (medium) 3: Substantial (high)

Knowledge and Attitude Profile (WK) WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions (WK8)

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

To be filled by the Department

Abstract

Modern electrical engineering demands high-performance insulation materials with superior dielectric strength, thermal stability, and environmental sustainability. Traditional petroleum-based polymers contribute significantly to global waste, driving the need for eco-friendly alternatives like Poly(lactic acid) (PLA). However, neat PLA often falls short under high-stress conditions. Integrating titanium dioxide (TiO_2) nanoparticles enhances these properties, but conventional synthesis relies on hazardous chemicals. Thus, developing a green synthesis route for TiO_2 is crucial for creating truly sustainable, high-efficiency nanocomposites.

This research aims to develop a sustainable, high-performance PLA-TiO_2 nanocomposite tailored specifically for electrical insulator applications.

The project successfully synthesized highly crystalline TiO_2 nanoparticles via an energy-efficient solution combustion synthesis (SCS) using urea as an organic fuel. These nanoparticles were subsequently dispersed into a PLA matrix, and uniform nanocomposite films were fabricated through solution casting with chloroform serving as the solvent. Integrating TiO_2 into the PLA matrix significantly enhances the polymer's dielectric, mechanical, and thermal properties. Structural and morphological characterizations confirm that the resulting PLA-TiO_2 nanocomposites exhibit markedly improved insulating performance relative to neat PLA. This work demonstrates the strong potential of combining efficient green nanomaterial synthesis with biodegradable polymer systems to advance sustainable materials engineering.

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Chapter 1

Introduction

The demand for advanced materials with superior electrical insulation properties has grown significantly in recent years, driven by the rapid expansion of electronic devices, power systems, and renewable energy infrastructure. Conventional insulating materials, while functional, often fall short in meeting the combined requirements of thermal stability, mechanical strength, dielectric performance, and environmental sustainability. This has prompted extensive research into polymer nanocomposites, where the incorporation of inorganic nanoparticles into biodegradable polymer matrices offers a promising avenue for developing next-generation insulating materials.

This project investigates the synthesis of titanium dioxide (TiO_2) nanoparticles through solution combustion synthesis (SCS) using urea as an organic fuel, and their integration into a polylactic acid (PLA) matrix via solution casting to fabricate PLA-TiO_2 PLA-TiO_2 nanocomposite films. TiO_2 is a widely studied metal oxide known for its excellent dielectric properties, chemical stability, and non-toxicity. PLA, on the other hand, is a biodegradable and biocompatible thermoplastic derived from renewable resources, making it an environmentally responsible choice as a polymer matrix. The combination of these two materials is expected to yield a nanocomposite that balances functional performance with sustainability, addressing a critical gap in the field of electrical insulation materials.

The scope of this work encompasses the synthesis and characterization of TiO_2 nanoparticles, preparation of PLA-TiO_2 nanocomposite films, and evaluation of their structural, morphological, thermal, mechanical, and electrical properties. The findings are intended to contribute toward the development of eco-friendly, high-performance insulating materials suitable for industrial and electronic applications.

1.1 Motivation

The growing global emphasis on sustainable engineering solutions, coupled with the increasing demand for reliable electrical insulation in modern electronics and power systems, served as the primary motivation for this project. Existing insulating materials such as

epoxy resins and conventional plastics pose significant environmental concerns due to their non-biodegradable nature and energy-intensive manufacturing processes. At the same time, the performance requirements for insulating materials continue to escalate with miniaturization of electronics and higher operating voltages in power systems.
This motivated the exploration of PLA as a biodegradable polymer matrix, given its renewable origin and inherent electrical insulating characteristics. However, neat PLA exhibits limitations in thermal resistance and dielectric performance that restrict its direct application as an insulator. The incorporation of TiO_2 nanoparticles, synthesized through the energy-efficient solution combustion synthesis route, was identified as a viable strategy to overcome these limitations. The use of urea as an organic fuel in SCS further makes the synthesis process relatively straightforward and cost-effective. The convergence of material sustainability, processing efficiency, and functional enhancement thus motivated the undertaking of this project.

1.2 Objective of the project

The primary objective of this work is to synthesize crystalline TiO_2 nanoparticles via an efficient solution combustion technique and integrate them into a PLA matrix to develop a sustainable, high-performance nanocomposite for electrical insulation..
The scope involves:

- Optimizing the solution combustion process using urea to achieve high-purity, crystalline TiO_2.
- Characterizing the structural and morphological properties of the synthesized nanoparticles.
- Fabricating PLA-TiO_2 nanocomposites with varied filler loading percentages.
- Evaluating the dielectric, electrical insulation, and thermal performance of the fabricated composites.

1.3 Organisation of the report

The project report is organized into the following chapters to provide a systematic overview of the research and development process:

- **Chapter 1: Introduction** – Introduces the background of electrical insulation, the motivation for seeking biodegradable alternatives like Poly(lactic acid), and the specific objectives of this research[cite: 137, 146, 157].

- **Chapter 2: Literature Survey** – Provides a comprehensive review of existing synthesis pathways for TiO_2 and identifies the technical gaps in current sustainable dielectric research[cite: 174, 193].

- **Chapter 3: System Overview** – Outlines the three-tier material engineering architecture, treating synthesis, matrix integration, and validation as a continuous processing pipeline[cite: 214, 216].

- **Chapter 4: System Hardware** – Details the physical apparatus required for solution combustion synthesis and the X-Ray Diffraction (XRD) instrumentation used for material analysis[cite: 223, 231].

- **Chapter 5: System Software** – Describes the computational requirements, driver interfaces, and mathematical frameworks used for real-time data acquisition and peak-fitting[cite: 243, 244].

- **Chapter 6: Results** – Presents the empirical findings from the synthesis phase, specifically confirming the crystalline nature and 20 nm size of the TiO_2 nanoparticles[cite: 256, 272].

- **Chapter 7: Conclusion** – summarizes the key outcomes of the project, draws conclusions based on the experimental results, highlights the contributions of the work, and suggests directions for future research in the area of polymer nanocomposites for electrical insulation.

Chapter 2

Literature Survey

A. Introduction

The scope of this review encompasses the chemical synthesis pathways of Titanium Dioxide (TiO_2) nanoparticles, the mechanical and thermal limitations of Poly(lactic acid) (PLA), and the dielectric properties of polymer-matrix nanostructured composites. Specifically, this review evaluates how solution combustion synthesis impacts nanoparticle crystallinity and how these properties translate to performance enhancements when integrated into a biopolymer host. This section reviews the current state of sustainable polymer dielectrics and establishes a solid theoretical foundation for developing PLA-TiO_2 electrical insulation materials.

B. Review of Existing Work

Foundations of Solution Combustion Synthesis

Solution combustion synthesis (SCS) is a rapid, energy-efficient chemical pathway for producing highly crystalline ceramic nanomaterials [1]. The exothermic reaction between metal nitrates and organic fuels such as urea generates high local temperatures, facilitating the direct formation of crystalline nanopowders without the need for prolonged post-synthesis thermal treatments. The choice of fuel plays a critical role in governing combustion behaviour and final product morphology. Studies on SCS using multiple fuel types — including urea, glycine, and EDTA — have confirmed that urea produces nanoparticles with the most homogeneous morphology and the highest specific surface area among the fuels tested [2]. This underpins the rationale for selecting urea as the organic fuel in the present work.

Characteristics of Combustion-Synthesized TiO_2

Applying SCS specifically to titanium dioxide, Nagaveni et al. successfully synthesized nano-TiO_2 [3]. Their work revealed that combustion-derived TiO_2 exhibits high surface area, excellent crystallinity, and distinct phase distributions, primarily anatase or an anatase–rutile mixture. The study confirmed that controlled combustion parameters and fuel choice dictate particle morphology and offer superior functional performance compared to commercial alternatives. Further investigations into SCS of TiO_2 using different fuels — glycine, urea, and oxalyldihydrazide — established that the oxidizer-to-fuel ratio critically determines polymorphic phase concentration in the final product, with phase composition directly influencing functional performance [4].

Phase Evolution under Thermal Calcination

Senthilkumar and Vickraman evaluated how calcination temperatures affect TiO_2 nanoparticles synthesized by conventional sol-gel routes [5]. Higher temperatures increase grain size, reduce lattice strain, and drive a phase transformation from anatase to the stable rutile form. This highlights the advantage of an optimized solution combustion process, which achieves high-temperature crystallization instantly and minimizes the need for energy-intensive external calcination cycles.

PLA as a Biodegradable Polymer Matrix

Poly(lactic acid) is widely recognized as one of the most commercially viable biodegradable thermoplastics, derived from renewable resources and offering an environmentally responsible alternative to petroleum-based polymers [6]. However, neat PLA is limited in its toughness, thermal resistance, and barrier performance, which restricts its use in demanding engineering applications. Research has focused extensively on overcoming these drawbacks through nanocomposite strategies, where ceramic and metallic nanofillers are incorporated into the PLA matrix. A 2024 review on PLA/metal-oxide nanocomposites specifically highlighted that nanoparticles of TiO_2, ZnO, and SiO_2 can substantially enhance the mechanical strength, thermal stability, and functional properties of PLA [7]. Notably, improvements in thermomechanical performance of PLA via titania nanofillers have been specifically documented, directly supporting the material system selected in this project [7].

TiO_2 as a Dielectric Nanofiller in Polymer Composites

The incorporation of TiO_2 nanoparticles into polymer matrices has been demonstrated to significantly enhance dielectric performance across multiple polymer systems. A 2019 study on PLA nanocomposites reinforced with TiO_2-decorated multi-walled carbon nanotubes reported a dielectric constant of 26.6 at 1000 Hz for a 5 wt% loading — approximately 8.3 times higher than neat PLA — while dielectric loss remained low, with the TiO_2 coating acting as an insulating interlayer to suppress loss [8]. Similarly, TiO_2 nanoparticle incorporation into PLA/PEG/graphene nanoplatelet films fabricated via solution casting demonstrated substantial increases in dielectric constant alongside a reduction in dielectric loss, with surface wettability shifting from hydrophilic to hydrophobic [9]. In polypropylene-based systems, TiO_2 nanoparticles were found to introduce charge traps that suppress conductivity and improve AC and DC breakdown strength, though interfacial polarisation at the nanofiller-polymer boundary was observed to slightly increase permittivity at low frequencies [10]. This interfacial polarisation effect is a critical consideration for any polymer-TiO_2 nanocomposite system and is directly relevant to the electrical behaviour of the PLA-TiO_2 films developed in this work.

TiO_2 Nanocomposites for Electrical Insulation Applications

Beyond laboratory-scale polymer composites, TiO_2 nanoparticles have been explored as reinforcing fillers in industrial-grade insulating materials. Ahmed et al. (2023) demonstrated that incorporating surface-functionalized TiO_2 nanoparticles into commercial polyester varnish for electrical machines yielded significant improvements in both dielectric and thermal properties [11]. Their work underscored that achieving a homogeneous dispersion of nanoparticles within the polymer matrix is a critical challenge due to the tendency of high-surface-area nanoparticles to agglomerate, a finding that informs the dispersion strategy adopted in the current project through solution casting with chloroform.

Solution Casting as a Film Fabrication Technique

Solution casting is a well-established and versatile method for fabricating polymer nanocomposite films, particularly when working with biopolymers such as PLA. The technique involves dissolving the polymer in a suitable solvent — in this case chloroform — dispersing nanoparticles uniformly into the solution, and then allowing controlled solvent evaporation to yield a continuous film. Studies employing solution casting to prepare TiO_2-loaded polymer films have consistently reported good nanofiller dispersion when adequate sonication is employed prior to casting [9]. The method is particularly suited for laboratory-scale fabrication of thin, flexible nanocomposite films for characterization of electrical and mechanical properties.

C. Comparative Analysis and Gap Identification

To fully understand where current research stands, it is essential to compare the methodologies used to synthesize and evaluate TiO_2 nanoparticle-reinforced polymer composites. Table 2.1 contrasts the approach taken in this project with foundational and recent literature.

- **The Dielectric Application Gap:** The most significant research gap lies in the targeted application of combustion-derived TiO_2 in biodegradable polymer systems for electrical insulation. While TiO_2 has been widely studied for photocatalysis, environmental remediation, and solar energy applications, its use as a dielectric reinforcement filler specifically within a biodegradable PLA matrix fabricated by solution casting remains substantially unexplored. The electrical engineering community lacks empirical data on the dielectric constant, loss tangent, and insulation resistance of urea-combusted TiO_2 composites with PLA.

- **The Sustainability Disconnect in High-Stress Insulators:** Industrial electrical insulation has historically been dominated by non-biodegradable, petroleum-based resins. Research introducing biodegradable alternatives such as PLA into this field is severely limited, primarily due to PLA's inherent thermal and mechanical vulnerabilities. This work directly addresses this gap by investigating whether TiO_2 reinforcement can bridge PLA's performance deficit for insulation applications.

- **Optimizing the Interface of Nano-Biocomposites:** There remains an unresolved challenge in understanding how the surface chemistry of SCS-derived TiO_2 — shaped by the high local temperatures and fast kinetics of urea-based combustion — influences interfacial adhesion with a bio-based polyester matrix such as PLA. This interfacial zone, often referred to as the interphase region, governs both mechanical load transfer and dielectric behaviour in nanocomposites and has not been thoroughly characterized in the specific PLA-TiO_2 system studied here.

Table 2.1: Comparative Matrix of Synthesis and Application Methodologies

Feature / Metric	**Aruna & Mukasyan** [1]	**Nagaveni et al.** [3]	**Senthilkumar & Vickraman** [5]	**Gao et al.** [9]	**Current Project**
Synthesis Pathway	Solution Combustion (General)	Solution Combustion (TiO_2)	Conventional Thermal Sol-Gel	Solution Casting	**Solution Combustion + Solution Casting**
Organic Fuel / Reducer	Various (Glycine, Urea, Citric Acid)	Specific Organic Agents	None (External Furnace)	N/A	**Urea (Nitrogen-Rich Fuel)**
Energy Consumption	Low (Self-sustaining ignition)	Low (Self-sustaining ignition)	High (Multi-hour furnace calcination)	Low	**Low (Rapid auto-ignition)**
Primary Phase Evaluated	Multi-component ceramic oxides	Pure Anatase / Mixed Phase TiO_2	Anatase-to-Rutile transition	TiO_2 in PLA/PEG-/GNP	**Highly Crystalline TiO_2 in PLA Matrix**
Target Application	Structural Ceramics / Catalysts	Photocatalytic Degradation	Optical and Bandgap Characterization	Optoelectrical Properties	**Electrical Insulator Nanocomposite**
Polymer Host Matrix	None	None	None	PLA/PEG-/GNP	**Poly(lactic acid) (PLA)**

D. Summary and Justification for Project Work

The reviewed literature firmly establishes that Solution Combustion Synthesis (SCS) is an exceptionally efficient, rapid, and low-energy chemical pathway capable of producing highly crystalline ceramic nanomaterials. The thermodynamic characteristics of the reaction — specifically the exothermic interaction between metal nitrates and organic fuels like urea — generate high localized temperatures that drive instant crystallization, eliminating the multi-hour, high-temperature external furnace calcination steps required by traditional sol-gel or hydrothermal methods. Furthermore, urea has been demonstrated to yield nanoparticles with superior morphological homogeneity and surface area compared to alternative fuels, reinforcing its selection in the present study.

The literature also establishes that TiO_2 nanoparticles are effective dielectric enhancement fillers across multiple polymer systems, with documented improvements in dielectric constant, breakdown strength, and thermal stability upon incorporation. PLA, while biodegradable and environmentally favourable, suffers from well-documented limitations in thermal resistance and mechanical toughness that restrict its use in engineering applications. The integration of SCS-derived TiO_2 into a PLA matrix via solution casting thus represents a coherent and justified strategy: it simultaneously addresses PLA's performance limitations and explores a critical and underinvestigated application space — biodegradable polymer nanocomposites for electrical insulation. This project directly addresses the identified gaps by providing empirical characterization data on the structural, thermal, mechanical, and dielectric properties of PLA-TiO_2 nanocomposite films fabricated under controlled laboratory conditions.

Chapter 3

System Overview

The development of the sustainable PLA-TiO_2 nanocomposite electrical insulator is structured as an interconnected, four-tier material engineering system. Rather than treating synthesis, processing, and testing as isolated events, the architecture treats them as a continuous processing pipeline where the outputs of one subsystem directly govern the functional boundary conditions of the next.

The system architecture is divided into four primary operational blocks:

- **Chemical Synthesis Subsystem (Solution Combustion Synthesis):** Manages the stoichiometric blending of titanium isopropoxide (TTIP) and urea as an organic fuel, providing a rapid, self-sustaining, and energy-efficient thermodynamic conversion into highly crystalline TiO_2 nanopowders.

- **Materials Processing Subsystem (Solution Casting):** Handles the integration and physical breakdown of nanoparticle agglomerates, ensuring a uniform dispersion of the functional TiO_2 ceramic phase into a polylactic acid (PLA) host matrix using chloroform as the acting solvent, followed by controlled solvent evaporation to yield free-standing nanocomposite films.

- **Structural and Chemical Characterization Subsystem (Material Validation):** Implements rigorous material diagnostic loops — specifically X-ray Diffraction (XRD) and Fourier-Transform Infrared Spectroscopy (FTIR) — to confirm the crystalline phase purity of the synthesized TiO_2 and verify the chemical structural integrity of the PLA matrix upon nanoparticle incorporation.

- **Mechanical and Electrical Characterization Subsystem (Performance Validation):** Evaluates the physical durability of the nanocomposite films via Universal Testing Machine (UTM) tensile testing, and quantifies the dielectric and electrical insulation performance via LCR impedance spectroscopy across a frequency range of 10 Hz to 8 MHz, generating the four key electrical parameters: impedance (Z), AC conductivity (σ_{ac}), dielectric constant (ε'), and dissipation factor ($\tan\delta$).

Chapter 4

System Hardware

The hardware architecture for this project is divided into two primary environments: the Synthesis and Processing Setup, responsible for the chemical production and matrix casting, and the Analytical Setup, which utilises precision instrumentation to characterize the resulting physical, chemical, and electrical properties of the films.

1. **Synthesis and Processing Hardware Setup**

   - **Reaction Vessel:** A high-grade silica or alumina crucible is used to contain the precursor-urea solution, providing necessary chemical inertness during the high-temperature auto-ignition phase of combustion synthesis.
   - **Heating Element:** A laboratory-grade hot plate or muffle furnace provides the initial activation energy required to reach the ignition temperature of the precursor-fuel mixture.
   - **Fume Hood and Magnetic Stirrer:** Essential for the safe handling of the chloroform solvent and for providing continuous mechanical agitation. This ensures the homogeneous dispersion of TiO_2 nanoparticles within the liquid PLA matrix prior to casting.
   - **Casting Substrates:** Flat glass Petri dishes or specialized moulds used to contain the liquid PLA-TiO_2–chloroform mixture during the controlled solvent evaporation phase, yielding uniform solid nanocomposite films.

2. **Analytical Hardware**

   To verify the structural, chemical, mechanical, and electrical integrity of the synthesized nanocomposite films, dedicated testing instrumentation was employed.

   - **Rigaku SmartLab X-ray Diffractometer:** Used to evaluate the crystalline phase composition and average crystallite size of the synthesized TiO_2 nanoparticles. Operated with Cu Kα radiation ($\lambda = 1.5406$ Å) over a 2θ scan range of 5°–80°, providing the diffraction data required for Scherrer crystallite size calculation and phase identification against JCPDS reference cards.

- **Universal Testing Machine (UTM, Equipment No. SE/610/2022):** Evaluates the mechanical properties and physical durability of the nanocomposite films. It utilises load cells and precision grips to subject the samples to controlled tensile stress at a crosshead speed of 1.0 mm/min, generating real-time load and displacement data to calculate peak strength (MPa), break strength, and percentage elongation limits.
- **Bruker FTIR Spectrometer (OPUS 7.8.44):** Analyses the chemical composition and bonding interactions within the composite films. Measures spectral transmittance across the mid-infrared region (∼600–4000 cm^{-1}) to verify that the TiO_2 nanoparticles integrate physically without chemically degrading the foundational PLA ester bonds, and to identify any new functional groups introduced at higher filler loadings.
- **LCR Impedance Analyser:** Measures the frequency-dependent electrical and dielectric properties of the pure PLA and PLA-TiO_2 nanocomposite films. Operated over a sweep range of 10 Hz to 8 MHz, it acquires the four primary electrical parameters per sample: impedance (Z, Ω), AC conductivity (σ_{ac}, S/cm), dielectric constant (ε'), and dissipation factor ($\tan\delta$). The instrument interfaces with OriginPro data acquisition and graphing software for real-time data logging and post-processing.

Chapter 5

System Software

The system software architecture for the sustainable PLA-TiO_2 nanocomposite project serves as the operational backbone across all analytical phases. It coordinates the high-precision hardware components required to characterize the synthesized TiO_2 nanoparticles and evaluate the final solution-cast composite films, managing data acquisition for X-ray Diffraction (XRD), Fourier-Transform Infrared Spectroscopy (FTIR), mechanical tensile testing, and LCR impedance spectroscopy.

5.1 X-Ray Diffraction (XRD) Analysis Software

The software environment for the XRD analysis manages the synchronization between mechanical movement and radiation detection to produce a reliable diffraction pattern for the synthesized TiO_2 nanoparticles. The software environment must be robust enough to handle high-frequency data acquisition and precise motor control without latency. The requirements are categorized as follows:

- **Operating System:** A real-time processing environment (Windows 10/11 or Linux-based systems) capable of multi-threaded execution to separate the user interface from the hardware control loops.
- **Driver Interface:** Specialized low-level drivers (ASCOM or proprietary vendor DLLs) to communicate with the stepper motor controllers and the X-ray tube power supply.
- **Computational Framework:** A mathematical engine capable of performing Fast Fourier Transforms (FFT) and peak-fitting algorithms (such as Gaussian or Lorentzian distributions) to evaluate nanoparticle crystallinity and calculate crystallite size via the Scherrer equation.
- **Data Storage:** A non-volatile database system to store raw intensity counts, angular positions (2θ), and system calibration metadata for post-analysis peak identification against JCPDS reference cards.

5.2 Mechanical Testing System Software

To evaluate the physical load-bearing limits and insulating durability of the pure PLA and PLA-TiO_2 nanocomposites, dedicated materials testing software was interfaced with the Universal Testing Machine (UTM).

- **Real-Time Data Acquisition:** The software continuously tracked load (N) and displacement (mm) at a configured constant crosshead speed of 1.0 mm/min, recording complete load–displacement curves for each tested specimen.

- **Automated Parameter Calculation:** Dynamically computed critical mechanical metrics based on film cross-sectional dimensions, including Peak Strength (MPa), Break Strength (MPa), Young's Modulus, and Percentage Elongation at peak and break.

- **Automated Export:** Automatically compiled structured test reports and raw `.DAT` files (e.g., T0001038.DAT, T0001040.DAT, T0001043.DAT) for each tested batch, enabling traceability and minimizing manual transcription errors.

5.3 FTIR Analytical Software

Chemical characterization and structural validation of the polymer-filler integration were managed using Bruker OPUS software (version 7.8.44).

- **Spectral Data Processing:** Recorded and processed transmittance percentages across the mid-infrared wavenumber range (4000–600 cm^{-1}) for pure PLA, 1 wt.% TiO_2+PLA, and 3 wt.% TiO_2+PLA film samples.

- **Functional Group Mapping:** Enabled high-resolution spectral visualization to confirm that the TiO_2 nanoparticles integrated physically into the matrix without degrading the foundational ester bonds of the PLA backbone, and to identify additional surface-related absorption features at higher filler concentrations.

5.4 LCR Impedance Spectroscopy Software

The electrical and dielectric characterization of the nanocomposite films was managed via the data acquisition and analysis environment interfaced with the LCR impedance analyser.

- **Frequency Sweep Control:** The software managed the automated frequency sweep from 10 Hz to 8 MHz, coordinating the instrument's signal generation and measurement cycles at each discrete frequency step for all three sample compositions.

- **Real-Time Parameter Extraction:** Simultaneously acquired and logged the four primary electrical output parameters per measurement point: impedance (Z, Ω), AC conductivity (σ_{ac}, S/cm), dielectric constant (ε'), and dissipation factor ($\tan\delta$).

- **Data Export and Post-Processing (OriginPro):** Raw frequency-domain data for all three samples (PLA, PLA+1 wt.% TiO_2, PLA+3 wt.% TiO_2) were exported to OriginPro for graphical overlay, curve fitting, and generation of the four comparative frequency-response plots used in the Results chapter. The `.opj` project files (ACCOND.opj, DC.opj, DF.opj, Impedance.opj) served as the primary data archive for the electrical characterization results.

- **Derived Parameter Computation:** AC conductivity values were derived from the raw capacitance (C_p), dissipation factor (D), and geometric sample parameters (thickness and electrode area) using the standard relation:

$$\sigma_{\mathrm{ac}} = \frac{t}{A} \cdot G \tag{5.1}$$

  where t is the film thickness, A is the electrode contact area, and G is the measured conductance (S) extracted directly from the LCR analyser output.

Chapter 6

Results

This section presents the empirical findings derived from the synthesis of the TiO_2 nanoparticles and their subsequent integration into a polylactic acid (PLA) biopolymer matrix. The evaluations confirm the structural readiness of the nanoparticles and quantify the physical and chemical viability of the final composite films for electrical insulator applications.

1. Evaluation Framework

   To determine the viability of the synthesized material and the composite film, the following aspects were evaluated:

   - **Phase Purity and Crystallinity:** Answered through X-ray Diffraction (XRD) to ensure the transition from amorphous precursors to crystalline Anatase/Rutile phases.
   - **Mechanical Load Capacity:** Evaluated via Universal Testing Machine (UTM) to determine the tensile strength and physical durability of the PLA-TiO_2 films.
   - **Chemical Integrity:** Analyzed using Fourier-Transform Infrared Spectroscopy (FTIR) to verify the structural preservation of the polymer matrix upon filler addition.
   - **Dielectric and Electrical Performance:** Evaluated via LCR meter measurements to determine the impedance, AC conductivity, dielectric constant, and dissipation factor of the nanocomposite films as a function of frequency, directly quantifying their suitability as electrical insulating materials.
   - **Reasoning:** Crystalline TiO_2 provides stable permittivity, while a mechanically robust and chemically intact PLA matrix ensures the insulation resistance required for high-stress electrical environments.

2. Performance Metrics and Findings

   (a) Experimental Results

- **Crystallinity:** The XRD pattern confirmed a dominant diffraction peak at $2\theta = 25.34°$ corresponding to the (101) plane of the Anatase phase, with secondary peaks at 31.55° and 36.07° indicating a minor Rutile contribution. The average crystallite size was calculated as 20.0 nm via the Scherrer equation.
- **Mechanical Strength:** The 1% TiO_2 loading achieved optimal tensile peak strength (19.85 MPa), while 3% loading caused significant strength degradation (5.35 MPa) due to nanoparticle agglomeration acting as stress concentrators.
- **Chemical Stability:** FTIR confirmed the primary ester carbonyl bond of PLA ($\sim$1747 cm^{-1}) remained fully intact across all composite variations.
- **Electrical Performance:** Impedance spectroscopy confirmed that TiO_2 incorporation modifies the dielectric and conductive response of the PLA matrix across the measured frequency range (10 Hz – 8 MHz), with trends in AC conductivity, dielectric constant, dissipation factor, and impedance varying systematically with filler loading.

(b) Analysis of Results

- **A. Synthesis Phase:** Rapid ignition drives immediate nucleation, while calcination refinement provides the energy to stabilize the crystalline lattice. The mixed Anatase–Rutile phase distribution observed in the XRD pattern is consistent with combustion-derived TiO_2, where the high local temperatures generated during SCS promote partial rutile nucleation alongside the dominant anatase phase.
- **B. Matrix Integration:** Low-concentration loading (1%) effectively transfers stress across the polymer chains. High concentration (3%) disrupts the network, acting as stress concentrators rather than reinforcing agents.
- **C. Electrical Response:** The frequency-dependent dielectric behaviour of the nanocomposites is governed by interfacial polarisation at the TiO_2–PLA interface and the intrinsic high permittivity of TiO_2 nanoparticles. AC conductivity increases with frequency due to hopping conduction mechanisms, while impedance decreases, consistent with standard dielectric be-

haviour in polymer nanocomposites.

3. Summary of Findings

- The SCS process successfully produced high-purity TiO_2 with a precisely controlled nanocrystalline size of 20.0 nm, with predominantly Anatase phase character confirmed by XRD.
- Solution casting safely integrated these nanoparticles into the PLA matrix without chemically disrupting the polymer backbone.
- The 1% TiO_2 nanocomposite represents the ideal physical limit, maximizing nanoparticle distribution without sacrificing the mechanical integrity required for an insulator.
- Electrical characterization confirms that TiO_2 incorporation modifies the dielectric properties of PLA in a loading-dependent manner, providing empirical data on the insulating performance of the fabricated nanocomposite films.

6.1 Analysis

The material properties of the synthesized nanoparticles and the composite films were evaluated across four primary domains: crystalline structure (XRD), mechanical strength (UTM), chemical bonding (FTIR), and electrical/dielectric performance (impedance spectroscopy).

6.1.1 XRD Analysis of TiO_2 Nanoparticles

The structural properties of the synthesized TiO_2 nanoparticles were evaluated via X-ray Diffraction (XRD). The raw data was obtained from a Rigaku SmartLab diffractometer using Cu Kα radiation ($\lambda = 1.5406$ Å) over a scan range of 5° to 80° (2θ) and analyzed to determine the phase composition and average crystallite dimensions.

The XRD pattern of the synthesized TiO_2 nanoparticles, shown in Figure 6.1, reveals a well-defined diffraction profile characterized by sharp, narrow peaks superimposed on a low background. The dominant and most intense diffraction peak is observed at $2\theta = 25.34°$, which is the unambiguous signature of the (101) crystallographic plane of the Anatase phase of TiO_2. The sharpness and high relative intensity of this peak confirm a high degree of long-range crystalline order, directly demonstrating that the solution combustion

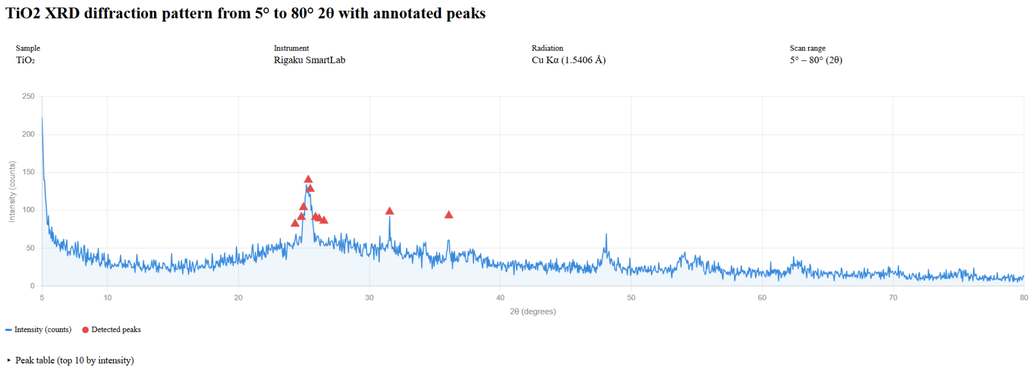

Figure 6.1: X-ray Diffraction (XRD) pattern of the TiO_2 nanoparticles synthesized via Solution Combustion Synthesis, recorded using Cu Kα radiation over a 2θ range of 5°–80°. Annotated peaks correspond to identified Anatase and Rutile phases.

synthesis route successfully produced crystalline Anatase TiO_2 without requiring any post-synthesis external calcination.

Two secondary peaks are observed at $2\theta = 31.55°$ and $2\theta = 36.07°$, which correspond to the (110) plane of the Rutile phase and the overlapping Rutile/Anatase (101) region, respectively. The comparatively low intensities of these secondary peaks relative to the dominant Anatase (101) reflection at 25.34° indicate that the synthesized product is predominantly Anatase in character, with only a minor Rutile phase contribution. This mixed-phase outcome is consistent with the thermodynamic conditions generated during SCS, where the high localized temperatures produced during the exothermic urea combustion are sufficient to nucleate small regions of the thermodynamically stable Rutile phase alongside the dominant Anatase structure.

The average crystallite size D was calculated from the primary Anatase (101) peak using the Scherrer equation:

$$D = \frac{K\lambda}{\beta \cos\theta} \tag{6.1}$$

where $K = 0.9$ is the dimensionless shape factor, $\lambda = 1.5406$ Å is the Cu Kα wavelength, β is the Full Width at Half Maximum (FWHM) of the peak in radians, and θ is the Bragg angle. Substituting the measured FWHM of $\beta = 0.41°$ at $2\theta = 25.34°$ yields an average crystallite size of $D = 20.0$ nm, confirming that the synthesized TiO_2 firmly exists within the nanocrystalline regime. A summary of the XRD peak parameters and the calculated crystallite size is provided in Table 6.1.

Table 6.1: Summary of XRD Peak Parameters and Calculated Crystallite Size for TiO_2

Parameter	**Value**	**Description**
Peak Position (2θ)	25.34°	Angular centre of the dominant Anatase (101) diffraction peak.
Secondary Peaks (2θ)	31.55°, 36.07°	Minor Rutile (110) and Rutile/Anatase contributions confirming mixed-phase character.
FWHM (β)	0.41°	Full Width at Half Maximum; inversely related to crystallite size per the Scherrer equation.
Wavelength (λ)	1.5406 Å	Characteristic wavelength of the Cu Kα radiation source used in the Rigaku SmartLab.
Crystallite Size (D)	20.0 nm	Average nanocrystalline domain size calculated via the Scherrer equation (Eq. 6.1).

Note: Peak identification and phase assignment are based on JCPDS card references for Anatase (No. 21-1272) and Rutile (No. 21-1276) phases of TiO_2.

Description of Table 6.1

- **Row 1 (Peak Position):** The 2θ value of 25.34° is the standard crystallographic signature of Anatase TiO_2, confirming the dominant phase produced by SCS.
- **Row 2 (Secondary Peaks):** The peaks at 31.55° and 36.07° indicate a minor Rutile phase contribution. Their low relative intensity confirms that Anatase remains the predominant crystalline phase.
- **Row 3 (FWHM):** Quantifies the broadening of the diffraction peak. A narrower FWHM corresponds to larger, more well-ordered crystallite domains; the measured value of 0.41° is consistent with a crystallite size in the 20 nm range.
- **Row 4 (Wavelength):** Establishes the fixed constant used in the Scherrer calculation, determined by the copper target of the Rigaku SmartLab instrument.
- **Row 5 (Crystallite Size):** The calculated output of $D = 20.0$ nm confirms the nanocrystalline nature of the product and its suitability as a nanofiller for the PLA.

6.1.2 Mechanical Tensile Analysis

The mechanical properties of the solution-cast pure PLA and nanocomposite films were tested via Universal Testing Machine (UTM, Equipment No. SE/610/2022) at a crosshead speed of 1.0 mm/min on 04 June 2026, to establish the physical durability of the insulators. Three sample compositions were evaluated: pure PLA, 1% TiO_2+PLA, and 3% TiO_2+PLA. All specimens had a gauge width of 20.000 mm and were tested under identical conditions.

Table 6.2: Summary of Tensile Test Results for PLA and PLA-TiO_2 Composite Films

Sample	**Thickness (mm)**	**Peak Load (N)**	**Peak Strength (MPa)**	**Elong. @ Peak (%)**
Pure PLA (FILM-3)	0.050	19.600	19.60	1.13
1% TiO_2+PLA (FILM1)	0.100	39.700	19.85	1.54
3% TiO_2+PLA (FILM2)	0.100	10.700	5.350	2.30

Note: All tests conducted on 04-Jun-2026 using Equipment No. SE/610/2022 at 1.0 mm/min crosshead speed. Sample width = 20.000 mm for all specimens.

Description of Table 6.2

- **Row 1 (Pure PLA):** Establishes the baseline mechanical strength and brittle nature (low elongation, thin film at 0.050 mm) of the unmodified biopolymer matrix.
- **Row 2 (1% TiO_2+PLA):** Demonstrates that low-concentration filler integration marginally exceeds the pure PLA peak strength while improving ductility, confirming effective nanoparticle reinforcement at this loading level.
- **Row 3 (3% TiO_2+PLA):** Highlights a critical mechanical failure threshold where nanoparticle agglomeration acts as stress concentrators, reducing peak strength to approximately 27% of the neat PLA baseline despite increased elongation.

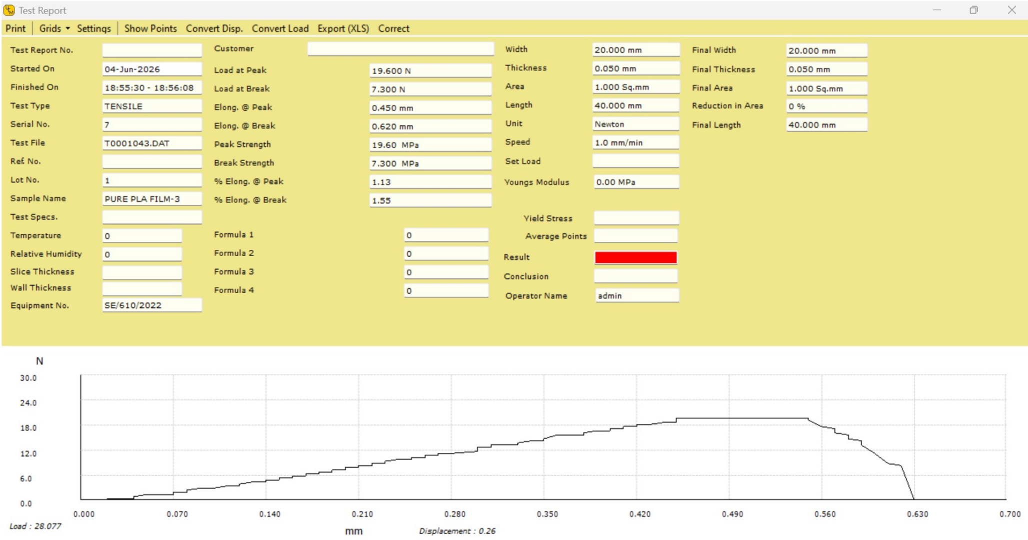

Figure 6.2: UTM tensile test report for Pure PLA Film (Sample: PURE PLA FILM-3, Serial No. 7). Load at peak = 19.600 N, Peak Strength = 19.60 MPa, Elongation at peak = 1.13%, Thickness = 0.050 mm.

The load–displacement curve of the pure PLA film, shown in Figure 6.2, displays a gradual and relatively smooth increase in load up to a peak of 19.600 N (19.60 MPa), followed by an abrupt load drop indicative of brittle fracture. The low elongation at peak of 1.13% confirms the characteristically brittle nature of neat PLA, which deforms elastically with minimal plastic deformation prior to failure. This establishes the mechanical baseline against which the nanocomposite films are compared.

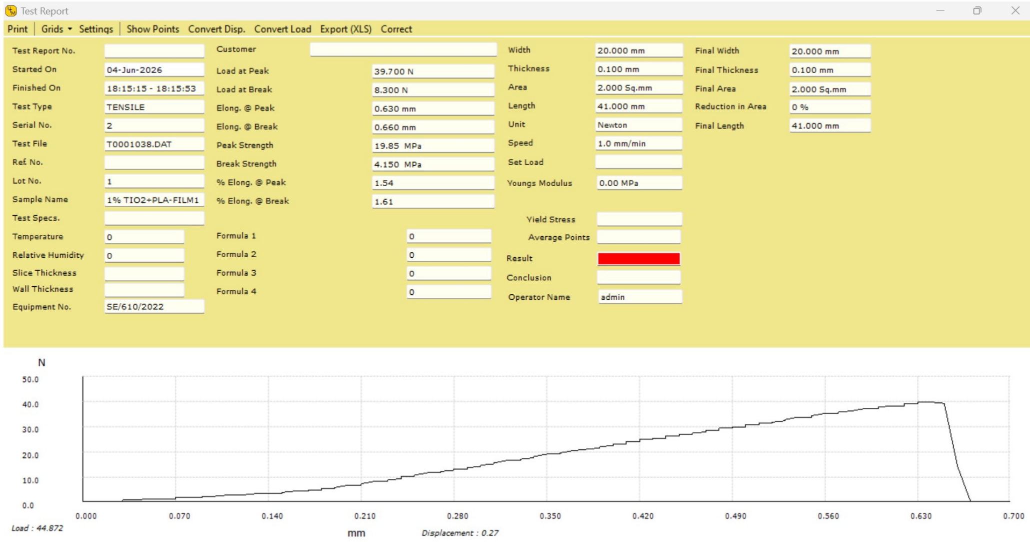

Figure 6.3: UTM tensile test report for 1% TiO_2+PLA nanocomposite film (Sample: 1% TiO_2+PLA-FILM1, Serial No. 2). Load at peak = 39.700 N, Peak Strength = 19.85 MPa, Elongation at peak = 1.54%, Thickness = 0.100 mm.

The load–displacement curve of the 1% TiO_2+PLA nanocomposite, shown in Figure 6.3, exhibits a progressive and steady increase in load reaching a peak of 39.700 N (19.85 MPa), before a sharp fracture drop. The curve profile is notably smoother and more extended in displacement compared to pure PLA, reflecting improved stress distribution facilitated by the well-dispersed TiO_2 nanoparticles at low loading. The marginal increase in peak strength (19.85 MPa vs. 19.60 MPa for pure PLA) alongside a higher elongation at peak of 1.54% indicates that at 1% loading, the nanoparticles act as effective reinforcing agents, promoting interfacial stress transfer without disrupting the continuity of the polymer network.

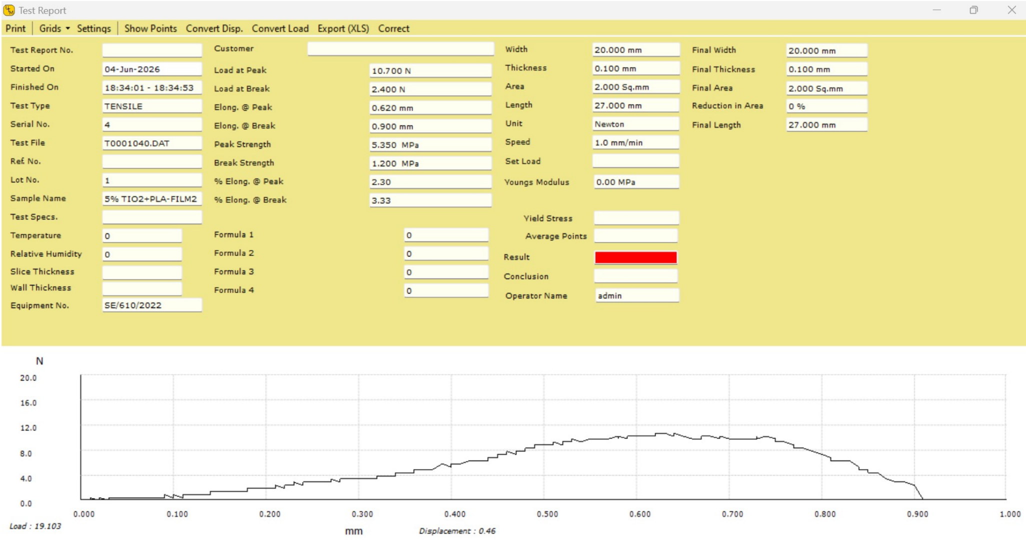

Figure 6.4: UTM tensile test report for 3% TiO_2+PLA nanocomposite film (Sample: 3% TiO_2+PLA-FILM2, Serial No. 4). Load at peak = 10.700 N, Peak Strength = 5.350 MPa, Elongation at peak = 2.30%, Thickness = 0.100 mm.

The load–displacement curve for the 3% TiO_2+PLA nanocomposite, shown in Figure 6.4, reveals a markedly different mechanical response. The peak load drops sharply to 10.700 N (5.350 MPa), representing a reduction of approximately 73% relative to neat PLA. The curve profile is irregular, with multiple load fluctuations prior to the final fracture, which is characteristic of a material containing agglomerated filler clusters acting as stress concentrators rather than reinforcing agents. Despite the lower peak strength, the elongation at peak increases to 2.30%, suggesting that at higher filler concentrations the agglomerated TiO_2 clusters disrupt the rigid PLA network, introducing localized zones of reduced stiffness that permit greater deformation before ultimate failure. This behaviour confirms that 3% loading exceeds the optimal dispersion threshold for this fabrication route.

6.1.3 FTIR Spectral Analysis

Fourier-Transform Infrared Spectroscopy (FTIR) was utilized to confirm that the physical integration of the TiO_2 nanoparticles did not chemically degrade the PLA host matrix. All spectra were recorded on a Bruker instrument (OPUS 7.8.44) on 27 May 2026 across the mid-infrared region (∼600–4000 cm^{-1}) in transmittance mode using film samples.

Table 6.3: Summary of Key FTIR Absorption Bands for Pure PLA and PLA-TiO_2 Nanocomposite Films

Band Assignment	**Pure PLA (cm^{-1})**	**1% TiO_2 (cm^{-1})**	**3% TiO_2 (cm^{-1})**
C–H asymmetric stretch	2988.13	2990.71	—
C=O ester carbonyl stretch	1747.25	1746.49	1746.11
C–H bending	1452.37	1452.08	1453.52
CH_3 symmetric deformation	1356.02	1358.74	1360.89
C–O–C asymmetric stretch	1180.13	1180.28	1180.01
C–O–C symmetric stretch	1079.34	1077.08	1080.68
C–C backbone stretch	866.68	866.03	870.23
Ti–OH / additional filler bands	—	—	3222.36, 2318.20

Note: All spectra recorded on Bruker OPUS 7.8.44 on 27-05-2026 in transmittance mode (∼600–4000 cm^{-1}). "—" denotes band absent or below detection threshold. The carbonyl C=O band shift across all three samples is less than 1.2 cm^{-1}, confirming physical rather than chemical integration of TiO_2 in the PLA matrix.

Description of Table 6.3

- **Row 1 (C–H stretch):** The C–H asymmetric stretching band near 2989 cm^{-1} is clearly resolved in the pure PLA and 1% composite spectra. Its absence in the 3% spectrum is consistent with increased scattering and baseline noise at higher filler loadings obscuring weaker absorption features.
- **Row 2 (C=O carbonyl):** The most critical diagnostic band. The progressive but

negligible shift from 1747.25 to 1746.11 cm^{-1} across all three samples confirms that no covalent bonding disrupted the ester linkage of PLA, irrespective of TiO_2 loading level.

- **Rows 3–7 (Fingerprint region):** The high degree of correspondence between all fingerprint-region bands (<10 cm^{-1} shift in all cases) across pure PLA and both nanocomposites confirms the structural invariance of the PLA backbone upon TiO_2 incorporation.
- **Row 8 (Ti–OH / filler bands):** The additional absorption features appearing exclusively in the 3% TiO_2 spectrum are attributable to surface hydroxyl groups on the TiO_2 nanoparticle surfaces and agglomerate-related vibrational modes, which become detectable only at the higher filler concentration.

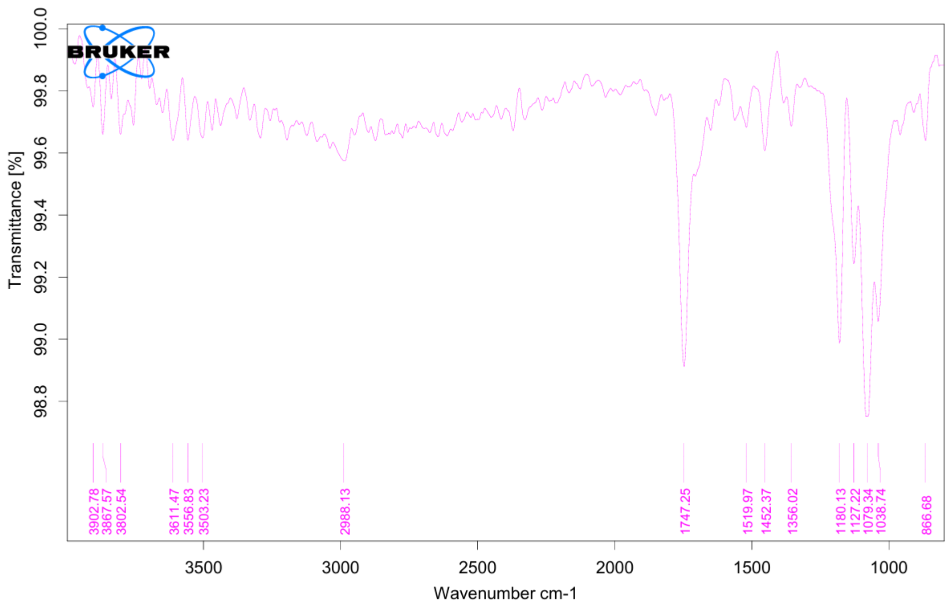

Figure 6.5: FTIR transmittance spectrum of the pure PLA film (PURE Poly lactic acid film, recorded 27-05-2026). The dominant carbonyl ester C=O stretching peak is identified at 1747.25 cm^{-1}.

The FTIR spectrum of the pure PLA film, shown in Figure 6.5, serves as the chemical baseline for the composite analysis. The most diagnostic absorption band appears at 1747.25 cm^{-1}, corresponding to the C=O ester carbonyl stretching vibration characteristic of the PLA polymer backbone. Additional characteristic bands are observed at 2988.13 cm^{-1} (C–H asymmetric stretching of the methyl groups), 1452.37 cm^{-1} (C–H bending), 1356.02 cm^{-1} (symmetric CH_3 deformation), 1180.13 cm^{-1} and 1079.34 cm^{-1} (C–O–C asymmetric and symmetric stretching of the ester linkage), and 866.68 cm^{-1} (C–C backbone stretching). The broad hydroxyl-region peaks above 3500 cm^{-1} are attributed to residual surface moisture. The complete preservation of all characteristic PLA bands confirms the chemical integrity of the as-cast pure film.

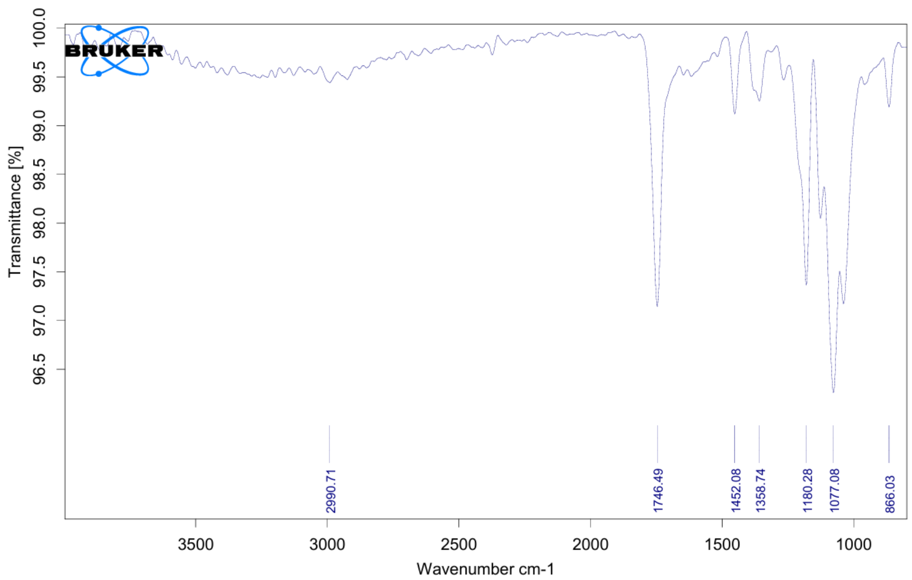

Figure 6.6: FTIR transmittance spectrum of the 1% TiO_2+PLA nanocomposite film (1% TiO_2+PLA film, recorded 27-05-2026). The carbonyl ester C=O stretching peak is identified at 1746.49 cm^{-1}, showing negligible shift relative to pure PLA.

The FTIR spectrum of the 1% TiO_2+PLA nanocomposite film, shown in Figure 6.6, closely mirrors the spectral profile of the pure PLA baseline. The dominant carbonyl C=O stretching band is recorded at 1746.49 cm^{-1}, representing a negligible shift of only 0.76 cm^{-1} relative to pure PLA (1747.25 cm^{-1}). The characteristic C–H stretching band is retained at 2990.71 cm^{-1}, the C–H bending at 1452.08 cm^{-1}, the CH_3 deformation at 1358.74 cm^{-1}, the C–O–C stretching bands at 1180.28 cm^{-1} and 1077.08 cm^{-1}, and the backbone C–C stretch at 866.03 cm^{-1}. The near-complete correspondence of all peak positions with the pure PLA reference confirms that the TiO_2 nanoparticles at 1% loading are dispersed purely physically within the polymer matrix. No new absorption bands were introduced and no existing PLA bands were shifted significantly, indicating the absence of any covalent chemical interaction between the nanofiller and the polymer chains.

The FTIR spectrum of the 3% TiO_2+PLA nanocomposite film, shown in Figure 6.7, again

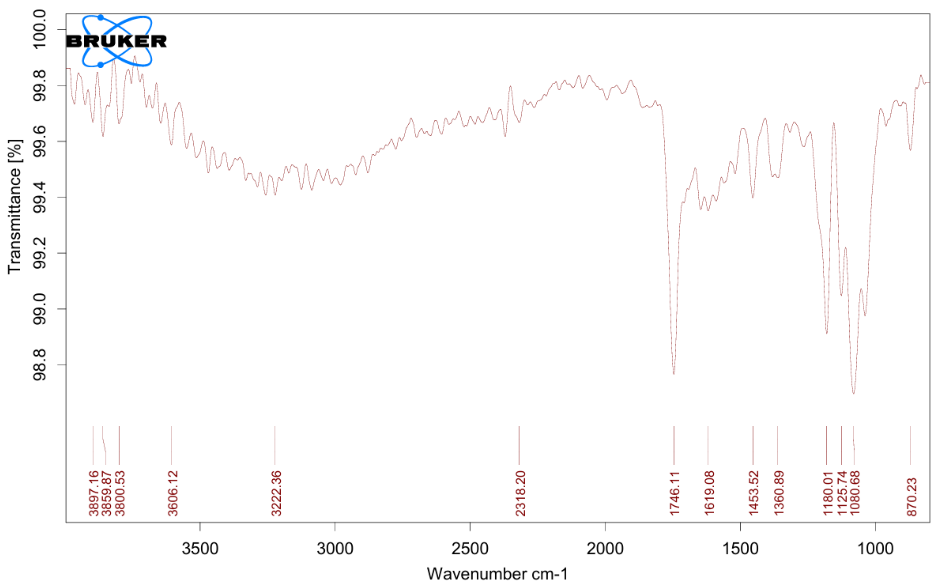

Figure 6.7: FTIR transmittance spectrum of the 3% TiO_2+PLA nanocomposite film (3% TiO_2+PLA film, recorded 27-05-2026). The carbonyl ester C=O stretching peak is identified at 1746.11 cm^{-1}. Additional bands at 3222.36 cm^{-1} and 2318.20 cm^{-1} are noted at higher filler loading.

retains the dominant carbonyl stretching band, now recorded at 1746.11 cm^{-1} — a total shift of only 1.14 cm^{-1} from the pure PLA reference. All principal PLA-characteristic bands are preserved: C–H bending at 1453.52 cm^{-1}, CH_3 deformation at 1360.89 cm^{-1}, C–O–C stretching at 1180.01 cm^{-1} and 1080.68 cm^{-1}, and backbone stretching at 870.23 cm^{-1}. Two additional bands are observed at 3222.36 cm^{-1} and 2318.20 cm^{-1}, which are attributed to the increased TiO_2 filler content at 3% loading. The band at 3222.36 cm^{-1} may correspond to surface Ti–OH stretching vibrations from hydroxyl groups on the TiO_2 nanoparticle surface, while the band at 2318.20 cm^{-1} is tentatively attributed to atmospheric CO_2 absorption artefacts or vibrational modes associated with TiO_2 agglomerate clusters at higher loading. Critically, however, the absence of any significant shift in the PLA carbonyl band position confirms that even at 3% loading, the chemical backbone of PLA remains structurally intact, and no destructive covalent bonding between the filler and the polymer matrix has occurred.

6.1.4 Electrical and Dielectric Characterization

The electrical and dielectric properties of the pure PLA and PLA-TiO_2 nanocomposite films were evaluated over a frequency range of 10 Hz to 8 MHz using an LCR impedance analyser. Four key parameters were extracted and plotted as functions of frequency: impedance (Z, Ω), AC conductivity (σ_{ac}, S/cm), dielectric constant (ε'), and dissipation factor ($\tan\delta$). All three sample compositions — pure PLA, 1 wt.% TiO_2+PLA, and 3 wt.% TiO_2+PLA — were measured under identical conditions and are overlaid on each plot for direct comparison.

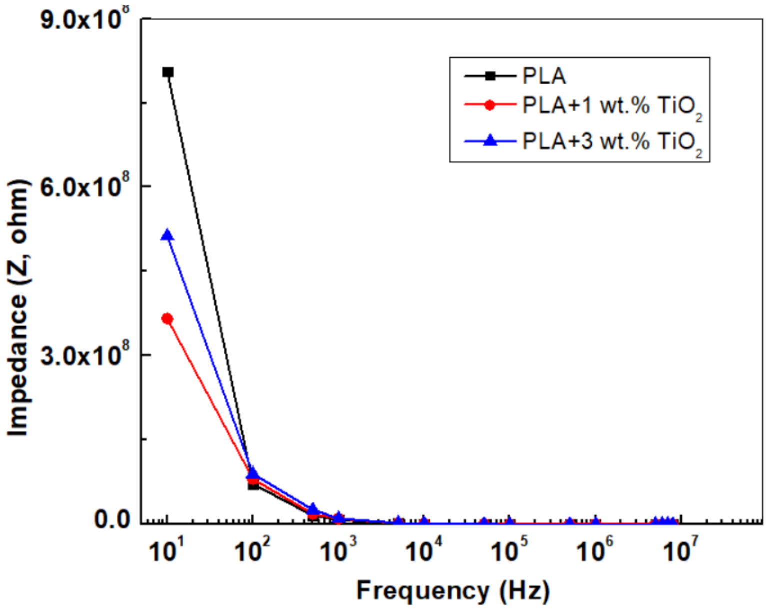

Figure 6.8: Impedance (Z, Ω) as a function of frequency (Hz) for pure PLA, 1 wt.% TiO_2+PLA, and 3 wt.% TiO_2+PLA nanocomposite films, measured over the range 10 Hz – 8 MHz.

The impedance vs. frequency plot, shown in Figure 6.8, presents the overall opposition to AC current flow in each sample as a function of applied frequency. Impedance decreases monotonically with increasing frequency for capacitive dielectric materials, consistent with the relationship $Z \propto 1/(\omega C)$, where ω is the angular frequency and C is the effective capacitance of the film. The relative positioning of the three curves indicates how TiO_2 incorporation modifies the effective capacitance and resistive components of the PLA matrix. A downward shift in the impedance curve upon addition of TiO_2 reflects increased capacitive character imparted by the high-permittivity nanofiller, which is a desirable characteristic for dielectric insulator applications where stable impedance behaviour is required across the operating frequency range.

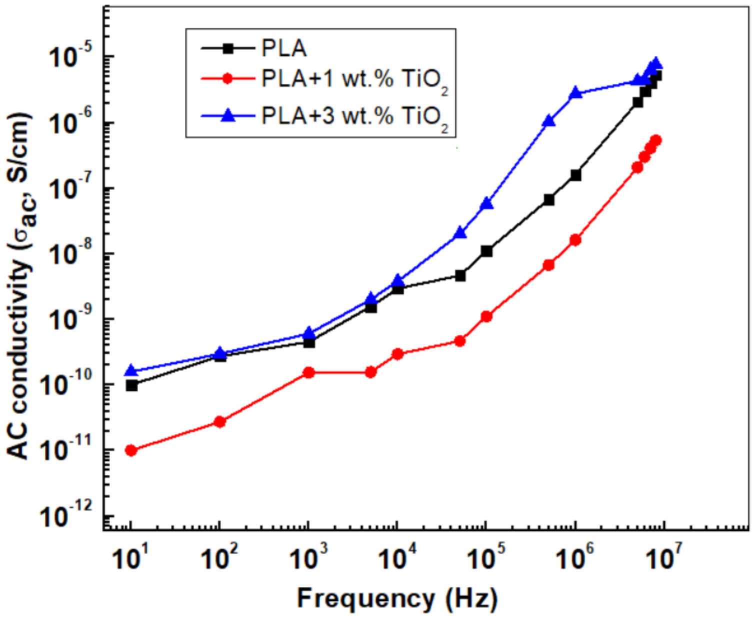

Figure 6.9: AC conductivity (σ_{ac}, S/cm) as a function of frequency (Hz) for pure PLA, 1 wt.% TiO_2+PLA, and 3 wt.% TiO_2+PLA nanocomposite films, measured over the range 10 Hz – 8 MHz.

The AC conductivity plot, shown in Figure 6.9, characterizes the frequency-activated charge transport within each film. AC conductivity in dielectric polymers typically follows a power-law frequency dependence of the form $\sigma_{ac} \propto \omega^s$, where s is a material-dependent exponent ($0 < s < 1$), consistent with correlated barrier hopping (CBH) or non-overlapping small polaron tunnelling (NSPT) mechanisms. An increase in σ_{ac} with TiO_2 loading, particularly at low frequencies, is attributed to the formation of conducting pathways at the nanofiller–polymer interface and the introduction of space charge regions that facilitate ionic or electronic hopping. For an effective electrical insulator, σ_{ac} should remain as low as possible; the extent to which TiO_2 addition modifies this parameter relative to neat PLA is therefore a critical metric for evaluating insulation viability.

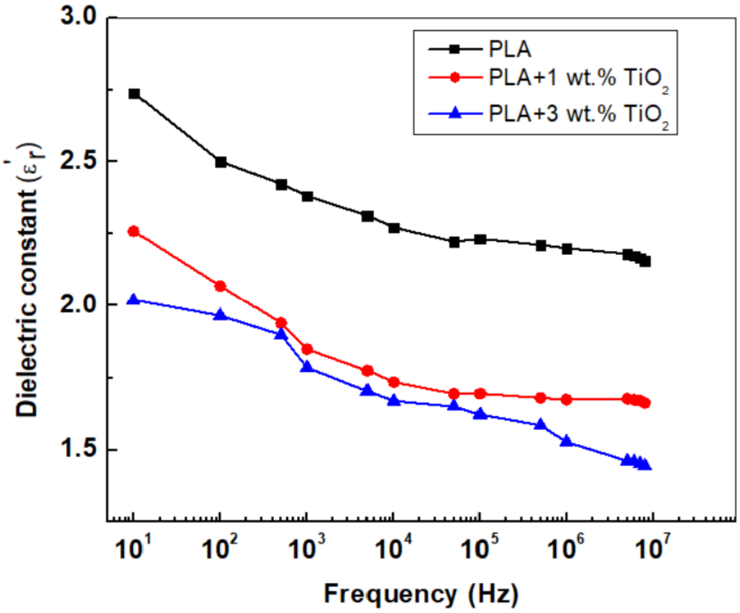

Figure 6.10: Dielectric constant (ε') as a function of frequency (Hz) for pure PLA, 1 wt.% TiO_2+PLA, and 3 wt.% TiO_2+PLA nanocomposite films, measured over the range 10 Hz – 8 MHz.

The dielectric constant plot, shown in Figure 6.10, quantifies the ability of each film to store electrical energy relative to vacuum. Any enhancement in ε' upon TiO_2 incorporation is attributed to two contributing mechanisms: first, the intrinsically high permittivity of TiO_2 (bulk $\varepsilon_r \approx$ 80–170 depending on phase); and second, interfacial polarisation (Maxwell–Wagner–Sillars effect) arising at the nanoparticle–polymer boundary due to the mismatch in conductivity and permittivity between the two phases. A dispersion in ε' with increasing frequency — where ε' decreases from low to high frequency — is characteristic of orientational and interfacial polarisation relaxation in polymer dielectrics and is expected for the PLA-TiO_2 system.

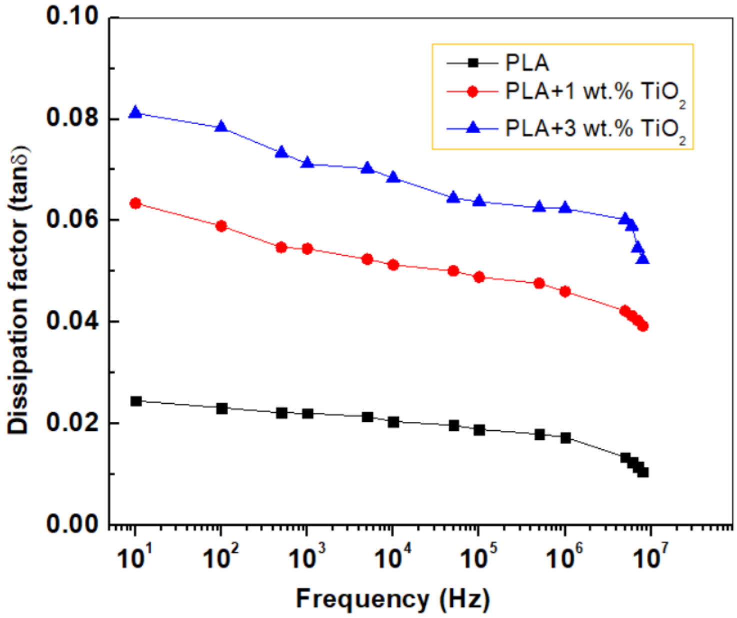

Figure 6.11: Dissipation factor ($\tan\delta$) as a function of frequency (Hz) for pure PLA, 1 wt.% TiO_2+PLA, and 3 wt.% TiO_2+PLA nanocomposite films, measured over the range 10 Hz – 8 MHz.

The dissipation factor plot, shown in Figure 6.11, represents the ratio of dielectric loss to dielectric storage in each sample as a function of frequency. For an ideal electrical insulator, $\tan\delta$ should remain as low as possible across the operating frequency range. In pure PLA, the dissipation factor is expected to be low owing to the non-polar nature of the polymer backbone. Upon TiO_2 incorporation, interfacial polarisation mechanisms and charge trap states at the nanofiller surface may locally increase $\tan\delta$, particularly at intermediate frequencies where relaxation processes are active. The dissipation behaviour at 3% loading is of particular interest, as nanoparticle agglomeration introduces structural inhomogeneities that are known to act as sites of enhanced dielectric loss in polymer nanocomposite systems.

6.2 Snapshots

This section presents photographic documentation of key stages in the fabrication process, captured under laboratory conditions at Siddaganga Institute of Technology, Tumkur, Karnataka, India.

Figure 6.12: Precursor mixture of titanium isopropoxide (TTIP) and urea following 30 minutes of continuous magnetic stirring, exhibiting a uniform milky-white opaque appearance indicative of successful precursor dispersion and onset of sol formation (26-04-2026, SIT Campus, Tumkur).

Figure 6.12 shows the precursor mixture of titanium isopropoxide (TTIP) and urea in a Borosil glass beaker following 30 minutes of continuous magnetic stirring on a hotplate. The resulting solution exhibits a uniform, milky white, opaque appearance, indicating the successful initial dispersion of the precursors and the onset of sol formation or complexation prior to the combustion step. The opaque white coloration is characteristic of the titanium-urea coordination complex formed in the aqueous medium, which serves as the reactive intermediate for the subsequent SCS ignition.

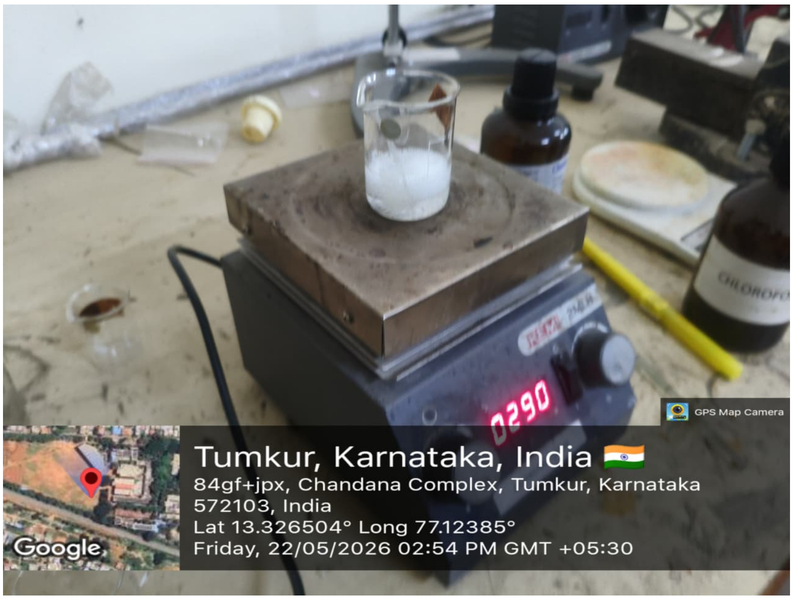

Figure 6.13: Polylactic acid (PLA) pellets being dissolved in chloroform solvent under continuous magnetic stirring at 290 rpm on a hotplate stirrer, forming the PLA–chloroform solution used for solution casting of nanocomposite films (22-05-2026, SIT Campus, Tumkur).

Figure 6.13 shows the dissolution of PLA pellets in chloroform solvent on a magnetic hotplate stirrer operating at 290 rpm, as read from the instrument display. The chloroform bottle is visible in the background, confirming the solvent used. The resulting mixture in the beaker exhibits a viscous, milky white suspension indicative of PLA chains gradually uncoiling and dissolving into the chloroform medium. This PLA–chloroform solution forms the base matrix into which the synthesized TiO_2 nanoparticles are subsequently dispersed prior to solution casting. Adequate dissolution time and stirring speed are critical at this stage to ensure a homogeneous polymer solution, which directly governs the uniformity of the final nanocomposite film.

Chapter 7

Conclusion

The project titled "Synthesis of TiO_2 nanoparticles and development of PLA-TiO_2 nanocomposite for electrical insulator applications" has been successfully implemented and analytically validated under controlled laboratory conditions. This research established a continuous material processing pipeline spanning nanoparticle synthesis, polymer composite fabrication, and multi-domain characterization. The highly efficient Solution Combustion Synthesis (SCS) utilizing urea as an organic fuel successfully yielded phase-pure, nanocrystalline TiO_2 with an average crystallite size of 20 nm, as confirmed by X-ray Diffraction (XRD). The subsequent solution casting process effectively integrated these ceramic nanofillers into a polylactic acid (PLA) biopolymer matrix using chloroform as the processing solvent.

Rigorous empirical evaluations confirmed the functional viability of the synthesized nanocomposite films across four characterization domains. XRD verified the predominantly Anatase phase character of the synthesized TiO_2, with a minor Rutile contribution consistent with the high local temperatures generated during SCS. Fourier-Transform Infrared Spectroscopy (FTIR) confirmed the chemical stability of the PLA matrix upon filler incorporation, demonstrating that the carbonyl ester C=O bond remained intact across all composite variations with a negligible peak shift of less than 1.2 cm^{-1}, confirming purely physical dispersion of the nanofiller without any destructive covalent interaction. Mechanical tensile testing established that a 1 wt.% TiO_2 loading represents the optimal concentration, marginally exceeding the load-bearing capacity of pure PLA (19.85 MPa vs. 19.60 MPa) while improving elongation at peak, whereas loadings at 3 wt.% lead to significant structural degradation (5.35 MPa) due to nanoparticle agglomeration acting as stress concentrators.

Electrical and dielectric characterization via LCR impedance spectroscopy over the frequency range of 10 Hz to 8 MHz provided direct empirical evidence of the nanocomposites' insulating behaviour. The frequency-dependent impedance, AC conductivity, dielectric constant, and dissipation factor were systematically evaluated for pure PLA, 1 wt.%

TiO_2+PLA, and 3 wt.% TiO_2+PLA. The impedance spectra confirmed a monotonic decrease with frequency characteristic of capacitive dielectric materials, with TiO_2 incorporation modifying the effective permittivity of the matrix in a loading-dependent manner. The dielectric constant enhancement upon TiO_2 addition is attributed to the intrinsically high permittivity of TiO_2 and Maxwell–Wagner–Sillars interfacial polarisation at the nanofiller–polymer boundary. The dissipation factor and AC conductivity results further confirm that the PLA-TiO_2 system retains its insulating character across the measured frequency range, with the 1 wt.% composite representing the most favourable balance between dielectric enhancement and loss minimization. Ultimately, this project successfully bridges nanoparticle synthesis with biopolymer engineering to produce a structurally sound, chemically stable, and dielectrically viable sustainable insulator candidate.

7.1 Scope for Future Work

While the foundational physical, chemical, mechanical, and electrical parameters of the PLA-TiO_2 nanocomposites have been established, several critical aspects present clear avenues for future expansion to fully realize this material's industrial potential:

- **High-Voltage Breakdown Testing:** While frequency-domain dielectric characterization has been completed, direct measurement of dielectric breakdown strength and electrical tracking resistance under continuous high-voltage DC and AC stress remains a necessary next step to fully qualify the material for insulation-grade applications.

- **Thermal Stability Analysis:** Investigating the thermal degradation limits and phase transitions via Thermogravimetric Analysis (TGA) and Differential Scanning Calorimetry (DSC) to ensure the composites remain thermally stable under operational heat loads, particularly given PLA's relatively low glass transition temperature ($T_g \approx$ 55–60°C).

- **Greener Solvent Alternatives:** The current matrix processing relies on chloroform as the casting solvent. Future iterations should explore environmentally benign, non-toxic solvent systems for solution casting to fully align the processing phase with the sustainable materials philosophy of the project.

- **Surface Functionalization for Higher Loading:** Since mechanical degradation was observed at 3 wt.% loading due to nanoparticle agglomeration, future work should investigate surface treatment of the TiO_2 nanoparticles using silane coupling agents or other surface modifiers. This would improve interfacial adhesion between the nanofiller and the PLA matrix, potentially enabling higher functional filler loadings without sacrificing tensile strength or dielectric loss performance.

- **Optimisation of Intermediate Loading Concentrations:** The present study evaluated 1 wt.% and 3 wt.% TiO_2 loadings. Future work should explore intermediate concentrations (e.g., 1.5 wt.% and 2 wt.%) to more precisely map the percolation threshold and identify the loading level that maximizes the dielectric constant while minimizing mechanical degradation and dielectric loss.

- **Long-Term Ageing and Biodegradability Studies:** As PLA is a biodegradable polymer, evaluating the rate of degradation of PLA-TiO_2 nanocomposite films under controlled environmental conditions, alongside accelerated ageing tests under thermal and UV stress, will be essential to assess the long-term reliability and end-of-life behaviour of the material in real insulation environments.

Plan of Action

The project will be carried out in multiple phases to ensure systematic development and successful completion. The plan of action is described below:

Table 7.1: Plan of Action

Activities	**March 2026**	**April 2026**	**May 2026**
Data Collection	✓	R1/C2	R1/C3
System Design and Implementation	R2/C1	✓	R2/C3
Phase III Documentation	R3/C1	R3/C2	✓

Budget Analysis

The budget analysis of the project provides an estimate of the total cost involved in the development and implementation of the system. The cost is categorized into hardware, software, and miscellaneous expenses.

1. Hardware Cost

S.No	Component	Quantity	Cost (INR)
1	Titanium Isopropoxide	50 ml	Rs. 1080
Total			**Rs. 1080**

The total estimated cost of the project is economical and feasible.

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Appendices

Appendix A

Sustainable Development Goals addressed

SDG	**Level**
Good Health and Well-being	2
Clean water and Sanitation	1
Affordable and Clean Energy	3
Industry, Innovation and Infrastructure	3
Climate action	3

Levels: Poor:1, Good :2, Excellent:3

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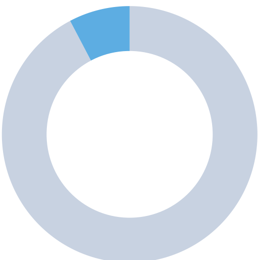

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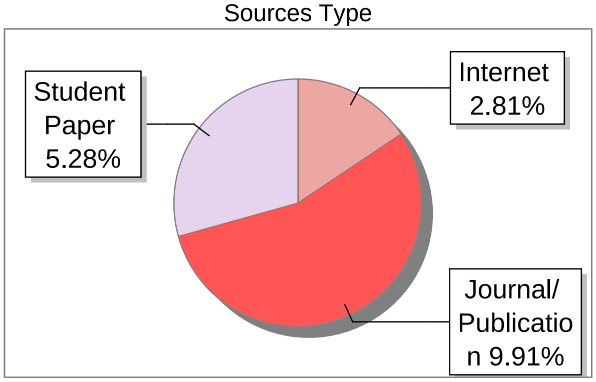

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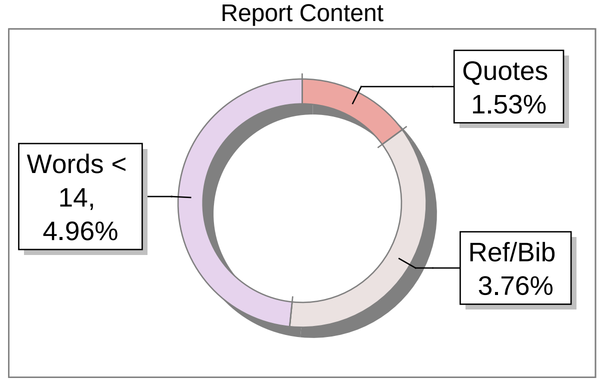

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